

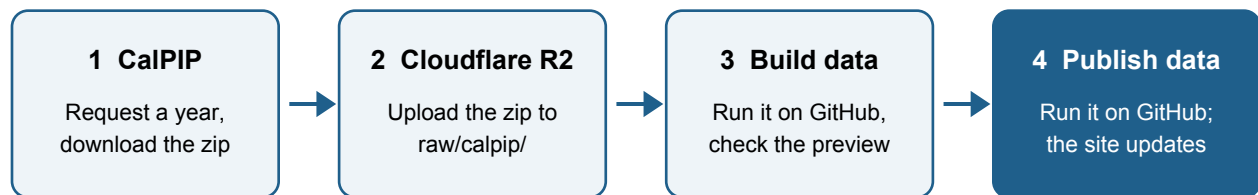
People and Pesticide Explorer: Updating the Data

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This guide replaces the March 2025 version. The explorer's data moved from Amazon Web Services (AWS) to Cloudflare R2 and GitHub Actions. Requesting data from CalPIP hasn't changed.

Adding a year of pesticide use data takes four steps:

- 1 Request the year from CalPIP and download the zip file. This takes about 20 minutes, mostly waiting for an email.
- 2 Upload the zip to the project's storage on Cloudflare R2.
- 3 Run the **Build data** job on GitHub and check the results. This takes about 10 minutes.
- 4 Run the **Publish data** job to put the new data on the live site.



Nothing changes on the live site until step 4, and step 4 can be undone at any time.

Before you start

- A Cloudflare login with access to the project's R2 bucket.
- A GitHub account with permission to run Actions in the **open-spatial-lab/cpr** repository.
- Dylan can set up both. Fill in these project details:

R2 bucket name: _____

Explorer website: _____

Key terms

- **CalPIP**: the website of the California Department of Pesticide Regulation (DPR) for downloading Pesticide Use Report (PUR) data.
- **Cloudflare R2**: online file storage, similar to Dropbox or Google Drive. The explorer's data lives here.
- **Bucket**: the top-level folder for a project in R2.

- **GitHub Actions:** buttons on GitHub that run scripts. *Build data* and *Publish data* are two of them.
- **Build:** a complete processed copy of the explorer's data, named by the date and time it was made, such as v20270115-0930. Builds are never changed or deleted, which is what makes undoing possible.
- **ZIP archive:** a compressed file, CalPIP's download format. Upload it as it is.

1. Request data from CalPIP

CalPIP (calpip.cdpr.ca.gov) is DPR's portal for pesticide use data. The explorer needs the full record-level data for the whole state: one row per pesticide application, not a summary. CalPIP allows one year per request, so repeat these steps for each year you're adding.

- 1 Open CalPIP and choose **PUR** as the data source. Click **Year** in the navigation menu and select the year you want. Leave the location, site, product and chemical filters empty.

The screenshot shows the CalPIP application interface. At the top is a blue header with the CA.gov logo, social media icons, and navigation links like 'About DPR', 'Join E-Lists', 'Careers', and 'Contact Us'. Below this is a white navigation bar with the DPR logo and links for 'About Pesticide Safety', 'SPM and IPM', 'Pesticide Registration', 'Professional Licensing', 'Data and Reports', and 'Get Involved with DPR'. A search bar is on the right.

The main content area is titled 'CALIFORNIA PESTICIDE INFORMATION PORTAL (CALPIP) APPLICATION'. It includes a 'Query Pesticide Use Report' section with a 'Select a Year' dropdown. The default selection is the most recent year of data. Below this, there's a list of 'Available Years (32):' from 2022 down to 2017. A note states: 'Note: Only one year may be selected per CalPIP query. Users who work with larger PUR datasets may download text files by year from the DPR File Site or contact PUR.Inquiry@cdpr.ca.gov'. At the bottom, there are 'My Selections' and 'Start Over' buttons.

On the left side, there's a 'CalPIP Navigation Menu' with links for 'CalPIP Application Home', 'My Selections', 'Data Sources' (PUR, GWPA, PRESCRIBE), 'Selected Data Source' (Pesticide Use Report), 'Search by Data Category' (Date, Year, Location, County, COMTRS, Zip Code, Site/Crop, Name Search, Product, Name Search, Chemical).

- Click **Format Output** in the navigation menu. Make sure **Summarize the Data** is **not** checked, and keep all output columns selected. The build checks for the columns it needs and will say if any are missing.

The screenshot shows the top navigation bar of the CalPIP website with the California Department of Pesticide Regulation logo and various menu items. Below the navigation bar, there is a section titled 'Summarize Data' with a checkbox labeled 'Summarize the Data'. Below this, there is a section titled 'Sort-By Columns' with a search box and buttons for 'Add All', 'Add Selected', 'Search', and 'Clear Search'. The search box contains the text '35 items displayed'.

- Under **Output File Format/Type**, keep **Tab-delimited text (default)** checked and **HTML table** unchecked. Click **Submit Query**.

The screenshot shows the 'Output File Format/Type' section of the CalPIP interface. It features a table with 4 items selected, including 'COUNTY_NAME', 'SITE_NAME', 'PRODUCT_NAME', and 'CHEMICAL_NAME'. Below the table, there is a section titled 'Output File Format/Type' with two checkboxes: 'Tab-delimited text (default)' (checked) and 'HTML table' (unchecked). At the bottom, there are buttons for 'Reset Page', 'My Selections', 'Start Over', and 'Submit Query'.

- 4 Enter your email address and click **Send Query Now**.

The screenshot shows the 'Query Pesticide Use Report' page on the CalPIP website. On the left is a 'CalPIP Navigation Menu' with links for 'CalPIP Application Home', 'My Selections', 'Data Sources' (PUR, GWPA, PRESCRIBE), 'Selected Data Source' (Pesticide Use Report), 'Search by Data Category' (Date: Year, Location: County, COMTRS, Zip Code, Site/Crop: Name Search, Product: Name Search, Chemical: Name Search, Other Criteria). The main content area is titled 'Query Pesticide Use Report' and contains instructions: 'Your query is ready to be sent to the database to extract data. You will be prompted for your email address if you have not entered it below. When your query has been run you will be notified by email with the URL to download the data file.' There is a text input field for 'Enter your email:' with a '(REQUIRED)' label below it. A red warning message states: 'Please review your email address for accuracy. If your address is entered incorrectly, we will be unable to notify you when your data results are ready.' Below this, it says 'You will be contacted through the email address you have entered strictly regarding your CalPIP query and not for any other purpose. You are responsible for picking up your data file. YOUR DATA FILE WILL BE DELETED IN 7 DAYS WHETHER IT IS CLAIMED OR NOT.' There is a link 'Click here to request CalPIP data:' and a 'Send Query Now' button. At the bottom, there are links for 'My Selections' and 'Start Over'.

- 5 CalPIP emails you a download link, usually within about 20 minutes. Download the zip file. Don't unzip it, rename the file inside it, or edit it. The link expires after 7 days.

2. Upload the zip to Cloudflare R2

The zip goes in the folder `raw/calpip/` in the project's bucket. That folder holds one file per year: zips that people have uploaded, and files named `calpip_<year>.parquet` for older years. Don't delete or rename those; every build uses all of them.

- 1 Log in at dash.cloudflare.com and choose the project's account.
- 2 In the left sidebar, open **R2 Object Storage**, then click the project's bucket.
- 3 Open the `raw` folder, then `calpip`.
- 4 Click **Upload** and choose the zip from your computer. Wait until the upload finishes before leaving the page.

Replacing a year. DPR sometimes revises past data. To re-pull a year, request it from CalPIP again, upload the new zip, and delete the old zip for that year if there is one. If both are in the folder, the build uses the one uploaded most recently.

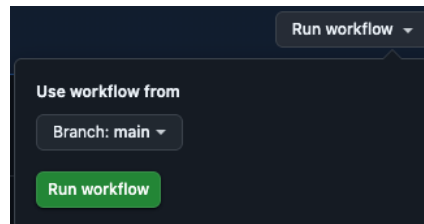
Large files

A full year from CalPIP is usually around 120 MB zipped, which the dashboard handles. Uploading through a browser can fail for much larger files; Cloudflare's own command-line uploader stops at 315 MB. If the upload fails, or the file is over 300 MB, use **Cyberduck**, a free app for Mac and Windows that splits large uploads into pieces automatically:

- 1 Install Cyberduck from cyberduck.io.
- 2 Get an access key (once). In the Cloudflare dashboard, open **R2 Object Storage** and click **Manage API tokens**. Create a token with **Object Read & Write** permission for the project's bucket only. Copy the **Access Key ID**, the **Secret Access Key**, and the S3 endpoint, which looks like `https://<account id>.r2.cloudflarestorage.com`. The secret is shown only once, so save it in a password manager. Dylan can also give you an upload key.
- 3 In Cyberduck, click **Open Connection** and choose **Amazon S3** from the list. For **Server**, enter the endpoint without `https://`, then the Access Key ID and Secret Access Key. Click **Connect**.
- 4 Open the bucket, then `raw` and `calpip`, and drag the zip into the window.

3. Build the data

- 1 Open [the Build data page](#) (in the repository, click **Actions**, then **Build data** in the list on the left).
- 2 Click **Run workflow**, then the green **Run workflow** button.



- 3 Wait for the run to finish, usually 5 to 10 minutes. Refresh the page: a green check means it worked, a red X means it stopped (see Troubleshooting).
- 4 Click the run to open its summary page and check three things:
 - **The version name and year range**, such as `v20270115-0930`, years 2017–2024. The year you added should be the last year.
 - **The yearly totals table**. The year you added shows as *new year*. Years already on the site should show 0.00% unless you re-pulled that year. A large change in an older year is a reason to stop and ask before publishing.
 - **The preview link**. It opens the explorer with this build's data and a banner saying it's a preview. Pick the new year in the date range and try a map and a chart.

Building doesn't change the live site. You can run Build data as many times as you like; the preview link always shows the most recent build.

4. Publish the data

- 1 Open [the Publish data page](#) (**Actions**, then **Publish data**).
- 2 Click **Run workflow**. Leave the **version** box empty to publish the most recent build, the one the preview link shows. Click the green **Run workflow** button.
- 3 It finishes in under a minute. Reload the explorer and it shows the new data. The website itself doesn't need to be rebuilt.

Undoing a publish. The Publish data summary page names the version it replaced. To go back, run Publish data again and type that version name into the **version** box. Every build is kept, so you can return to any of them.

Troubleshooting

Problem	What to do
Build data shows a red X, and the error mentions <i>missing CalPIP columns</i> .	The zip is a summary or is missing columns. Request the year again (section 1, steps 2 and 3), replace the zip, and run Build data again.
The year you added isn't in the totals table.	The zip isn't in <code>raw/calpip/</code> , or it was unzipped before uploading. Check the folder and that the file name ends in <code>.zip</code> .
Build data fails with a different error.	Send Dylan the link to the failed run. The live site isn't affected.
The CalPIP download link has expired.	Submit the CalPIP query again.
The upload in the Cloudflare dashboard fails.	Use Cyberduck (section 2, Large files).
The site shows the wrong data after publishing.	Run Publish data with the previous version (section 4, Undoing a publish).

For maintainers: the pipeline code and setup are described in the repository's README.